Lesson 1 Computer Vision and OpenCV Introduction

1. How robots “see” the world

For artificial intelligence, the ability to see is essential. And how robots see the world involves machine vision, an important branch of artificial intelligence.

Machine vision is the idea that the robot takes human’s place to measure and make judgments. The captured target will be converted into image signal by image sensor, CMOS or CCD, and then the image signal will be transferred to the specialized image processing system which will convert the image signal into digitized signal according to the pixel distribution, brightness, color, etc.

Image system perform various operations on these signals to extract the features of the target, so as to control the device in the field based on the judgments.

Machine vision technology is commonly applied in intelligent transport system and intelligent housing system.

2. Image Recognition Introduction

Image recognition is a crucial technique that uses computer to process and analyze the image so as to recognize different targets.

Similar to human eyes, machine image recognition starts at the point where there is huge variance or sudden change, and the features will recognized one by one. Our brain controls our eyes to collect the major features of the image and filter the redundant information, and then integrate the major features into the complete visual image.

The process of computer image recognition is no different from that of human image recognition, which is divided into four steps.

1) Acquire information: the light signal, sound signal, etc., are converted

into electric signal by the sensors to acquire the information

2) Image preprocessing: perform smoothing, denoising, etc., on the image to highlight the major features of the image.

3) Feature extracting and selecting: extract and select the image features, which is the pivotal step.

4) Image classification: make the recognition rules that is design classifier based on the training result to get the main category of the features so as to improve the recognition accuracy.

Image recognition is mostly applied in remote sensing image recognition and robot vision.

3.OpenCV Introduction

OpenCV (Open Source Capture Vision) is a computer vision library for free handling various tasks about image and video, for example display the image collected by the camera and make the robot recognize the real object.

OpenCV is more eminent than PIL, the built-in image processing library in Python. OpenCV provides complete Python interfaces, and Python3.5 and opencv-python library file have been integrated in the provided image system.

4. How Images are Stored in Computer

How the images are stored in computer after they are recognized?

In general, a picture is composed of pixels and each pixel can be represented by R, G and B components within 0-255. OpenCV stores each pixel as a ternary array making it convenient to record all information of the image. In addition, OpenCV records the data of three color channels of RGB image in the order of BGR.

Besides, images of other standards (HSV) are stored as multivariate array. An OpenCV image is a two-dimensional or three-dimensional array. An 8-bit grayscale image (black-and-white images) is a two-dimensional array, and a 24-bit BGR image is a three-dimensional array.

For a BGR image, the first value of the element “**image[0,0,0]**” represents Y--axis coordinate or the row number(0 represents the top). The second value represents X-axis coordinate or column number (0 represents leftmost). And the third value represents the color channel.

Same as Python array, these array recording images can be accessed individually to obtain the data of some color channel or capture a region of the image.

Lesson 2 Build OpenCV Environment

1. Install Numpy

Each picture involves several pixels, which results in that a large number of arrays need to be processed in the program. Numpy is a extension library for Python, which handles multi-dimensional arrays more efficiently than Python's native array structures. Besides, it can improve the readability of codes.

Open command line terminal and then input command "**pip install numpy**" to install Numpy. For more information about Numpy, please move to the folder "**3.Python->Python Basic and Advanced Learning->Lesson 13 Python Numpy Basic Operation**".

2. Install OpenCV

OpenCV package can be obtained from Ubuntu repository. Then refresh the packages index and install the OpenCV package by typing the following commands.

1) sudo apt update: refresh the packages index

2) sudo apt install python3-opencv: Install the package. During installation, input "y" to continue the execution and the complete installation may take 10s.

```
ubuntu@ubuntu-virtual-machine: ~
File Edit View Search Terminal Help
ubuntu@ubuntu-virtual-machine:~$ sudo apt install python3-opencv
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

3. Verify the Installation of OpenCV

We can import cv2 module to print the version of OpenCV so as to verify whether the installation is successful or not.

1) python3: enter Python

2) import cv2: import cv2 module

3) cv2.__version__: check the version

If the version of OpenCV is printed, the installation is successful.

```
ubuntu@ubuntu-virtual-machine:~$ python3
Python 3.6.9 (default, Jan 26 2021, 15:33:00)
[GCC 8.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import cv2
>>> cv2.__version__
'3.2.0'
```

Lesson 3 OpenCV Modules and Components

1.OpenCV Component

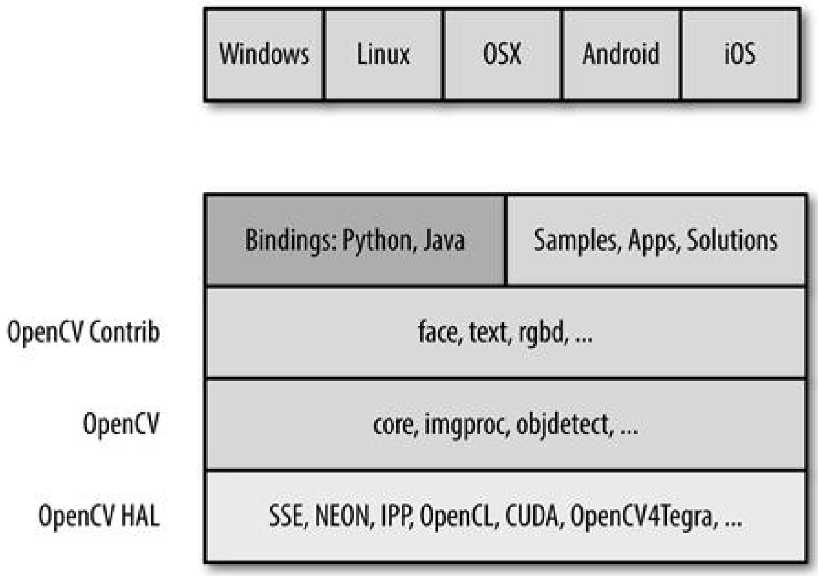
OpenCV is composed of several layers of modules.

1) The bottom layer is the hardware optimization based on HAL (Hardware Acceleration Layer)

2) Above the bottom layer are the codes contributed by other developers contained in opencv_contrib module. These codes, core of OpenCV, involves most of the high-level functionality.

3) The next layer are language bindings and sample applications.

4) The top layer is the interaction between OpenCV and operating system.



2.Specific Module of OpenCV

1) Core: Contain the basic structure and operation of OpenCV library.

2) Improc: Image processing module can transform the basic image, including filtering and convolution.

3) Highgui: Seen as lightweight Windows UI Toolkit, it is divided into imcodecs, videoio and highgui in OpenCV 3.0. It contains user-interaction function used to display the images or simple input.

4) Video: Contain the functions for reading and writing the video streams.

5) Calib3d: Contain the algorithm of the calibration of single, binocular and multiple cameras.

6) Feature2d: Used for the algorithm of feature point detection, description and matching.

7) Objdectect: Contain the algorithm of specific target detection, including human face or passengers. And it can be used to train the detector to detect other objects.

8) Ml: Machine learning module is a comprehensive module that involves a mass of machine learning algorithms which can interact with OpenCV data type.

9) Flann: Flann stands for Fast Library for Approximate Nearest Neighbors, which will be called by the functions in other modules for fast nearest neighbor search in large datasets.

10) GPU: It is segmented into several cuda* modules in OpenCV. GPU module can optimize the functions on CUDA GPU and involves the functions only applicable to GPU. Without GPU, the computing resources cannot be promoted causing that some functions cannot return good results.

11) Photo: A new module that contains the functions of computational photography.

12) Stitching: Also a new module that stitches sophisticated images

13) Nonfree: It is moved to opencv_contrib/xfeatures2d in OpenCV 3.0. There are some algorithms that is protected by patent and limited in usage in

OpenCV, such as SIFT. These algorithms are isolated into their own modules, therefore you need to take special measures to use them in commercial products.

14)Contrib: It involves something new that haven't been integrated into OpenCV.

15)Legacy: It has been removed from OpenCV 3.0. This module contains some old stuffs that haven't been completely removed.

16)Ocl: Khronos OpenCL standard. It has been removed from OpenCV 3.0 and replaced by T-API. Similar to GPU module, it realizes Khronos OpenCL standard for open parallel programming.

Compared with GPU module, it has fewer functions, but it aims at providing the parallel devices that can run on any GPU or is powered by Khronos. However, GPU module can only run on Nvidia GPU devices for the reason that it utilizes Nvidia CUDA toolkit to develop.

Lesson 4 Picture & Video Loading and Display

1.Image Reading and Writing

Read image: cv2.imread(Location，Model)

1) Location——read the location of the image which can be the absolute path and relative path. However pay attention to the usage of the slash in different operating system.

2) Model——model of image loading. The first model is cv2.IMREAD_COLOR used to load a color picture but will not load Alpha channel(record degree of transparency). The second one is cv2.IMREAD_GRAYSCALE which is used to load a grayscale picture. The third type is cv2.IMREAD_UNCHANGED for loading image and Alpha channel simultaneously.

3) Display image: cv2.imshow("Name", Pic)

4) Name——Display the box name of the image

Pic——Pictures to be displayed(The image read by cv2.imread() has already used before) For example, create a new py file and put the picture named “**camera.png**” into the same folder. Then input the following codes. After the codes run, the image will be displayed and you can press any key to hide the image.

```
import cv2
a = cv2.imread("camera.png")
cv2.imshow("test", a)
cv2.waitKey()
cv2.destroyAllWindows()
```

Note: cv2 waitkey() allows users to display a window for given milliseconds or until any key is pressed. And cv2.destroyALLWindows() function will close all the windows.

2.Video Reading and Writing

Video can be seen as pictures that are switched swiftly. Therefore, video reading is the extension of the image reading and writing. Camera initialization: cv2.VideoCapture(Number)

1) Number——Serial number of camera, 0 usually.

Read the frame of camera：cap.read()。

2) cap——the camera that has been defined before

Release the resources of the camera：cap.release()。

For example, the camera screen will be displayed on the desktop. When q key is pressed, the camera screen will be hidden.

```
import cv2
cap = cv2.VideoCapture(0)
while(cap.isOpened()):
    ret,frame=cap.read()
    cv2.imshow('capture', frame)
    key = cv2.waitKey(1)
    if key &0x00FF==ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

Note: cv2.waitKey(delay) will wait for the input from the keyboard and can be used to refresh the image in the video. “**delay**” in the bracket indicates the waiting time. When a frame of picture is displayed, the program will display the next frame in “delay” ms.

Lesson 5 Image Drawing

Drawing function in OpenCV can be used to draw line, rectangle, circle, etc., and add texts to the designated position of the picture.

1.Draw Line

Function format: **cv2.line(image,pt1,pt2,color,thickness)**

1) Image: Image where the line will be drawn

2) pt1: starting coordinate of the line. The coordinate is represented by a tuples consisting of two values i.e. (X,Y)

3) pt2: ending coordinate of the line. The coordinate is represented by a tuples consisting of two values i.e. (X,Y).

4) Color: The color of the line. And BGR is represented by a tuple. For example, (255, 0, 0) stands for blue.

5) Thickness: The thickness of the line

```
cv2.line(image,(100,100),(100,200),(255,0,0),5)
```

2.Draw Rectangle

Function format: **cv2.rectangle(image,pt1,pt2,color,thickness)**

1) image: The picture where the rectangle will be drawn

2) pt1: vertex coordinate of the rectangle, (x,y), which is represented by a tuple consisting of two numbers.

3) pt2: The diagonal vertex coordinates of pt1 and its format is similar to that of pt1.

4) color: The color of the rectangle. And BGR is represented by a tuple.

For example, (255, 0, 0) stands for blue.

5) thickness: Line thickness. The greater the value, the thicker the line. If the value is negative or cv2.FILLED, a filled rectangle will be drawn.

```
cv2.rectangle(image,(100,100),(200,200),(255,0,0),5)
```

.

3. Draw Circle

Function format: **cv2.circle(image,center,radius,color,thickness)**

1) image: The picture where the circle will be drawn

2) center: The center of the circle, (x,y), which is represented by a tuple consisting of two numbers.

3) radius: The radius of the circle.

4) color: The color of the circle. BGR is represented by a tuple. For example, (255, 0, 0) stands for blue.

5) thickness: Line thickness. The greater the value, the thicker the line. If the value is negative or cv2.FILLED, a filled circle will be drawn.

```
cv2.circle(image,(100,100),50,(255,0,0),5)
```

4. Draw Polygon

Function format: **cv2.polylines(image,pts,isClosed,color,thickness)**

1) image: The picture where the polygon will be drawn

2) pts: The vertex coordinate of the polygon. When several quadrangles are required in a picture, the shape of ndarray is (N，4，2).

3) isClosed: Whether the polygon is closed or not, True generally.

4) color: The color of the polygon. BGR is represented by a tuple. For example, (255, 0, 0) stands for blue.

5) thickness: Line thickness. The greater the value, the thicker the line.

```
pts=np.array([[10,10],[100,10],[100,100],[10,100]],np.int32)
pts=pts.reshape((-1,1,2))
cv2.polylines(image,[pts],(255,0,0),5)
```

5. Add Text

Function format: **cv2.putText(image,text,pt,font,fontScale,color)**

1) image: The image where the text is added.

2) text: The text content

3) pt: The coordinate of the upper left corner of the text

4) font: Font of the text

5) fontScale: Font size

6) color: The color of the text. BGR is represented by a tuple. For example, (255, 0, 0) stands for blue.

```
cv2.putText(image,"Hello World!",(100,100),font,5,(255,0,0))
```

Lesson 6 Image Basic Operation

1.Acquire and Modify the Pixel of the Image

The value of the pixel can be acquired through the coordinate of row and column. For BGR image, an array consisting of blue, green and red values will be returned. For grayscale image, the corresponding intensity will be returned. And the pixel can be modified in this way.

1) **img[x,y]**: Acquire the value of some pixel and return its BGR value.

2) **img[x,y,index]**: Acquire the value of a color channel. The order of the color channel is BGR.

3) **img[x,y]=[B,G,R]**: Modify the color channel value of this pixel.

```
img=cv2.imread("test.jpg")
px=img[100,100]
blue=img[100,100,0]
img[100,100]=[255,255,255]
```

2.Acquire the Image Property

1) **shape**: If it is a color picture, acquire the shape of the image and return an array containing the number of row, column and channel. If it is binary image or grayscale image, only the number of row and column will be returned. Through judging whether the returned value contains the number of channel, we can know that it is a grayscale picture or color picture.

2) **size**: Return the pixel number of the image. The format is "**row x column x channel**". The number of channel of the grayscale picture is 1.

3) **dtype**: Return the data type of the picture

```
print("shape=",img.shape)#"(1600,1200,3)"
print("size=",img.size)#"5760000"
print("dtype=",img.dtype)#"uint8"
```

3. Splitting and Merging of Image Channel

3.1Splitting of Image Channel

split: Input the image to be split and return the picture with three individual color channels.

```
img=cv2.imread("test.jpg")
B,G,R=cv2.split(img)
cv2.imshow("blue",B)
cv2.imshow("green",G)
cv2.imshow("red",R)
```

3.2 Merging of Image Channel

merge: Merge three individual channels, including B, G and R into BGR image with three channel.

```
img=cv2.imread("test.jpg")
B,G,R=cv2.split(img)
img=cv2.merge((B,G,R))
```

4. Color Space Conversion

There more than 150 ways to convert colors in OpenCV. And BGR is commonly converted into GRAY and HSV. The function format is **cvtColor(img,flag).**

1) **img**: The image converted the color space

2) **flag**: The converted type. For example, **cv2.COLOR_BGR2HSV** indicates that convert BGR into HSV.

```
img=cv2.imread("test.jpg")
img=cv2.cvtColor(img,cv2.COLOR_BGR2HSV)
```